MADHUMANTI SAHA

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LinkedIn GitHub

PROFESSIONAL SUMMARY

Results-oriented Electrical Engineering undergraduate (B.Tech, 2026) from Academy of Technology, Eager and dedicated seeking an Associate Engineer Trainee role. Possess strong fundamentals in electrical systems, hands-on industrial exposure, and experience in IoT-based, data-driven and AI Automated projects. Looking to contribute effectively in manufacturing, operations, and engineering environments.

EDUCATION

B.Tech – Electrical Engineering (EE) | Academy of Technology, MAKAUT 2022 – 2026

Current CGPA: 7.29/10 (Up to 7th Semester)

Class XII (WBCHSE) – Udaypur Haradayal Nag Adarsha Vidyalaya 2020 – 2022

Aggregate: 77.6%

Class X (WBBSE) – Bagbazar Multipurpose Girls' School 2015 – 2020

Aggregate: 81.8%

TECHNICAL SKILLS

- **Core Concepts:** Electrical Machines, Motors & Drive Characteristics, Control Systems, Power Systems
- **Tools & Technologies:** MATLAB, Simulink, Power BI, MS Excel, Raspberry Pi 4B/Pico W, AutoCAD (Basics)
- **Programming Languages:** Python, Java
- **Data & ML Libraries:** SQL, NumPy, Pandas, Scikit-learn, Matplotlib, Power BI

PROJECTS

DC Shunt Generator Load Characteristic Analysis via IoT [GitHub]

Raspberry Pi 4B/Pico W | MQTT | React | Scikit Learn | Matplotlib

- Built a full-stack IoT system to remotely monitor and control a DC Shunt Generator using MQTT protocol, WebSockets, and a React.js web dashboard.
- Applied Signal Processing **(EMD)** and statistical analysis on motor drive datasets to identify electrical and mechanical anomalies, establishing groundwork for ML-based **predictive maintenance**.
- Engineered an Adaptive Control mechanism using PWM signals to regulate prime mover speed remotely, maintaining stable output under varying industrial load conditions.

Customer Churn Analysis & Prediction Pipeline [GitHub]

Python | Pandas | Scikit-learn | Power BI | SQL | Power Query

- Built a complete **end-to-end data pipeline** covering ingestion, cleaning, EDA, and feature engineering to identify key churn drivers from large datasets.
- Developed and evaluated ML classification models (Logistic Regression, Random Forest) using **ROC-AUC, precision, and recall** for data-driven decision prioritisation.
- Created **Power BI dashboards** with segmentation visualisations to communicate actionable insights to stakeholders.

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- **Industrial Vocational Training – Eastern Railway (Liluah)** – Completed hands-on training in carriage and wagon maintenance, including inspection, repair of mechanical and electrical systems, wheel-set alignment, braking systems, bogie overhauling, and industrial safety practices. [Verify]
- **Smart India Hackathon (SIH) 2024** – Recognised among the **Top 8 teams nationally** for an innovative engineering solution.
- **RC Car Competition Winner** – Led a 5-member cross-functional team to first place in a college-level RC Car Racing event, demonstrating team coordination and strategic planning.
- **IEEE Student Member** – Active member of IEEE Student Branch (AOT); participates in technical seminars and workshops.